TEXORA · LECTURE SHEET

Foam Technology in Wet Processing

Replacing most of the water in a finishing or dyeing liquor with air — how stable foam is made, measured, applied and collapsed on the fabric, and why low wet pick-up saves so much drying energy.

WP-4203 · SPECIAL WET PROCESSING · TOPIC 1 · SPECIALIZATION — WET PROCESSING

1. What foam processing is — and why it matters

Foam technology in wet processing means dispersing a process liquor as a fine, metastable foam — a dispersion of air bubbles in a small amount of liquid containing surfactant — and applying that foam to fabric instead of immersing or padding the fabric in the full liquor. The foam is a *carrier*: it delivers the dissolved or dispersed chemicals (dye, resin, softener, flame retardant, water repellent) uniformly over the fabric surface, then collapses on or in the fabric, releasing its small liquor content exactly where it is needed.

The driving motive is water and energy. A conventional pad applies 60–80% wet pick-up; most of that water does no chemical work — it only has to be evaporated again in the stenter. A foam application typically applies 15–35% wet pick-up. Less water carried means less water to dry, less chemical migration during drying, shorter drying sections, lower effluent volume and — in foam dyeing — less hydrolysis of reactive dyes. Foam processing is therefore classed as a **low-liquor, low wet pick-up (LWPU)** technology, alongside spray, kiss-roll and minimum-application systems.

Foam is *not* the product. A good processing foam must do three contradictory things in sequence: (1) be stable enough to survive transport from generator to applicator without draining or coarsening; (2) spread uniformly over a moving fabric; and (3) collapse promptly and completely on the fabric so that its liquor is released evenly, not in spots. Every formulation and machine choice in this sheet is a negotiation between stability and controlled collapse.

2. Foam generation and stability

2.1 How foam is made

Industrial foam is generated mechanically. Liquor containing a **foaming agent** (a surfactant such as an alkyl sulphate, alkyl ether sulphate or a commercial foam concentrate) and usually a **foam stabiliser** (a long-chain fatty alcohol, a polymeric thickener or a protein-type stabiliser) is metered together with

compressed air into a **foam generator** — a mixing head in which a rotor–stator, a packed bed of beads or mesh, or a sintered element shears the air into fine bubbles. Bubble diameters of about 50–200 µm are typical; fine, uniform bubbles drain slowly and coat evenly, whereas coarse, polydisperse foam drains fast and applies unevenly.

A surfactant lowers the surface tension of the liquor and adsorbs at each air–water interface, forming an elastic film (the **Gibbs–Marangoni effect**: stretching a film locally depletes surfactant, raises local surface tension and pulls liquid back, healing the thin spot). The films separating two bubbles form a **lamella**; where three lamellae meet, a channel called a **Plateau border** forms, and it is through these borders that liquid drains under gravity and capillary suction.

2.2 Why foam dies — drainage, coarsening and rupture

- **Drainage:** liquid flows out of lamellae into Plateau borders and downward. The foam becomes drier at the top, wetter at the bottom. Drainage rate rises with bubble size and falls with liquor viscosity — hence stabilisers and thickeners.
- **Coarsening (Ostwald ripening / disproportionation):** gas diffuses from small bubbles (higher internal Laplace pressure, $\Delta P = 2\gamma/r$ for a single surface) to large bubbles. Small bubbles shrink and vanish; large ones grow; the foam coarsens even without drainage.
- **Film rupture:** dirt, oil, silicone defoamer contamination, hard-water ions or mechanical shock thin a lamella below its critical thickness and it bursts, merging neighbours.

Stability is quantified practically by the **foam half-life** (time for half the liquid to drain, or for the foam column to fall to half height, in a standard cylinder test) and by drainage curves. A finishing foam needs a half-life comfortably longer than the residence time between generator and fabric — typically minutes, not seconds — but not so long that it survives on the fabric and causes resist marks. Temperature, pH, electrolytes and dissolved dyes all shift stability, so the foam must be tested with the *actual* process liquor, never with water alone.

3. Blow ratio and foam density — the two control numbers

The **blow ratio** is the volume ratio of foam produced to liquor consumed:

$$\text{Blow ratio} = V_{\text{foam}} / V_{\text{liquor}} \quad | \quad \text{Foam density } \rho_f \approx \rho_{\text{liquor}} / \text{blow ratio}$$

Blow ratios in textile foam work are typically 5:1 to 20:1. A high blow ratio gives a dry, light foam and very low wet pick-up, but the foam is more fragile and harder to spread; a low blow ratio gives a wet, robust foam closer to ordinary padding. Foam density is measured continuously in production (weighing a known foam volume) because it directly fixes the application: *mass of liquor applied per unit area = foam density × foam layer thickness*.

WORKED EXAMPLE 1 — FOAM DENSITY FROM BLOW RATIO

Liquor density ≈ 1000 g/L. Blow ratio 10:1.

Foam density = 1000 / 10 = **100 g/L**.

Meaning: 1 litre of this foam contains only 100 mL of liquor and about 900 mL of air. If the applicator lays a uniform foam layer that transfers, say, 250 mL of foam per kg of fabric, the liquor actually delivered is 250 × 0.100 g/mL = **25 g per kg = 2.5% liquor by foam volume accounting** — in practice the controlled variable is the resulting wet pick-up (Section 5), measured gravimetrically on the fabric.

Control loop in a modern line: liquor flow meter + air flow meter set the blow ratio; an in-line density cell trims the air valve; applicator gap or knife pressure sets the layer thickness. If density drifts, shade and finish level drift with it — density, not generator speed alone, is the quality variable.

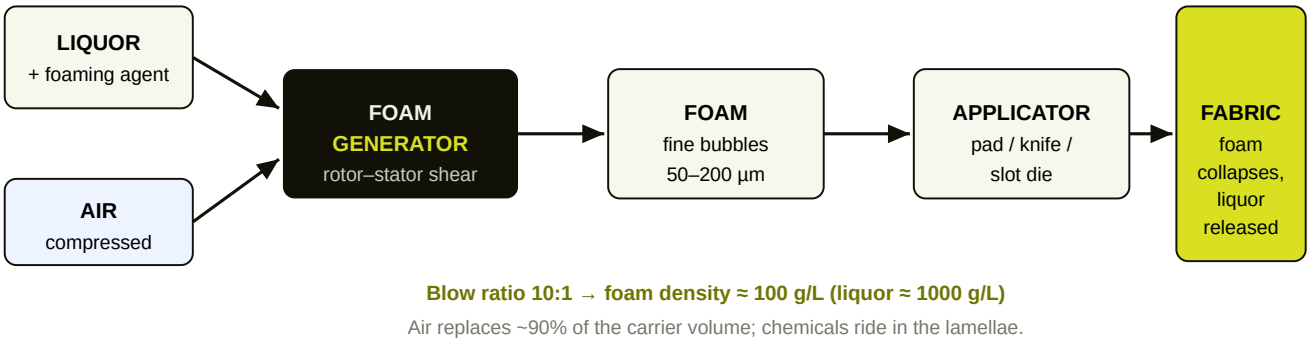


Figure 1 — Foam generation and application line. Liquor and air are sheared in the generator; the foam is metered by the applicator and collapses on the fabric.

4. Foam application systems

SYSTEM	HOW IT WORKS	TYPICAL USE
Horizontal pad (foam pad)	Foam is fed into the nip of a horizontal padder (or onto the fabric just before the nip). Squeezing crushes the foam and forces liquor in. Closest retrofit to conventional padding.	Finishing, resin application, one-side effects with vacuum control
Knife-over-roll / knife-over-table (blade)	A doctor blade spreads a foam blanket of set thickness over fabric supported by a roll, table or blanket. The foam collapses under the blade and by capillary absorption. Excellent for one-sided application.	Coating-type finishes, foam finishing of pile and upholstery fabrics
Foam finishing (general)	Any of the above used to apply chemical finishes — softeners, easy-care resins, water/oil repellents, flame	Energy saving, reduced migration, two-sided

	retardants, antimicrobials — at 15–35% pick-up instead of 60–80%.	differential finishes
Foam dyeing	Dye liquor foamed and applied uniformly, then fixed by steaming, thermosoling or pad-batch dwell. Low liquor limits reactive-dye hydrolysis; suits continuous and carpet dyeing.	Carpets, pile fabrics, denim effects, pigment colouration
Foam printing	Coloured foam applied through a rotary/flat screen or by patterned crushing. Air replaces much of the thickener paste; print definition depends on foam stability under the squeegee.	Blotch and effect prints, pile printing
Foam coating / lamination assist	Foamed adhesive or coating compound spread and collapsed/cured as a porous layer.	Back-coating, interlining, flocking adhesive

Common line elements: generator → distribution manifold (foam must be fed *evenly across the full width* — a manifold with balanced path lengths, because foam ages as it travels) → applicator → (optional vacuum slot to pull liquor through pile) → dryer/stenter → curing. Two practical rules: keep the generator-to-applicator distance short (foam ages, drains and coarsens in the pipe), and never pump finished foam through a high-shear pump — use low-shear displacement or gravity feed, or the bubbles you paid to make are destroyed before use.

5. Low wet pick-up — advantages quantified

Wet pick-up (WPU) = (wet mass − dry mass) / dry mass × 100%. Reducing WPU attacks the largest single energy item in finishing: evaporating water. Only the latent heat is counted in the simple comparison below (real dryers also heat fabric, air and machine, so absolute fuel savings are smaller in percentage terms, but the evaporation term dominates and scales directly with WPU).

WORKED EXAMPLE 2 — DRYING-ENERGY SAVING (LATENT HEAT 2260 KJ/KG)

Conventional pad, WPU 70%: water carried = 0.70 kg per kg fabric.
Evaporation energy = 0.70 × 2260 = **1582 kJ per kg fabric**.
Foam application, WPU 25%: water carried = 0.25 kg per kg fabric.
Evaporation energy = 0.25 × 2260 = **565 kJ per kg fabric**.
Saving in evaporation energy = (1582 − 565) / 1582 ≈ **64%**.
At line scale this also means a shorter dryer can reach the same speed, or the same dryer can run faster — the energy saving converts directly into capacity.

- **Less migration:** with little free water, dissolved chemicals cannot migrate to yarn cross-over points or fabric edges during drying — more level finishes and fewer edge-to-centre differences.
- **One-sided and differential effects:** a collapsed foam layer wets mainly the face it touches; face/back differences (soft face, firm back; repellent face, absorbent back) become possible in one pass.
- **Less hydrolysis / higher fixation:** in foam dyeing with reactive dyes, less water means less dye lost to hydrolysis.
- **Lower effluent:** less liquor prepared, less left to drain at the end of a run; foam residues collapse to a small liquid volume.
- **Handle:** less water + less drying stress often preserves bulk and softness, valuable for pile, knits and nonwovens.

6. Foam finishing recipes — concept

A foam recipe is written like a pad recipe but with two extra functional ingredients and a foam specification. Conceptual easy-care foam finish (for illustration of structure, not a production order):

COMPONENT	FUNCTION	NOTES
Crosslinking resin (e.g. modified DMDHEU type)	Active finish	Dosed to deliver the same g/m² as the pad recipe — recalculate from target WPU, do not copy g/L blindly
$MgCl_2$ catalyst	Curing catalyst	Ratio to resin as supplier specifies
Softener (silicone or fatty type)	Handle	Must be foam-compatible; some silicones defoam — check
Foaming agent	Creates the foam	Chosen for stability window and easy rinsing/collapse
Stabiliser / thickener	Slows drainage	Raises lamella viscosity; excess harms penetration
Water	Carrier	Balance to 1 L of liquor before foaming

Recipe conversion rule: chemical delivered per kg fabric = liquor concentration × WPU. If a pad recipe delivers resin at 80 g/L and 70% WPU, delivery = 80 × 0.70 = 56 g/kg. To deliver the same 56 g/kg at 25% WPU, the foam liquor needs 56 / 0.25 = **224 g/L** resin. Foam liquors are therefore far more concentrated than pad liquors — solubility, viscosity and compatibility must all be re-checked at the higher concentration, and the foam specification (blow ratio, density, half-life) is written on the recipe sheet beside the chemistry.

7. Limitations and foam collapse control

- **Penetration:** foam sits on the surface; dense, hydrophobic or tightly woven fabrics may show poor penetration and a surface-only finish. Remedies: lower blow ratio (wetter foam), vacuum assist, pre-wetting, or accepting foam only for suitable substrates.
- **Uniformity risk:** any density or distribution drift prints itself onto the fabric as streaks. Width-wise distribution and density control are the make-or-break engineering.
- **Compatibility:** many textile chemicals defoam (silicone softeners, oils, some pigments) or destabilise (high electrolyte, extreme pH). Every new recipe needs a foam test with the full formulation.
- **Collapse control:** collapse on the fabric is triggered by (i) mechanical crushing (nip, blade, vacuum), (ii) capillary suction into the fabric, and (iii) chemical defoaming on contact. If foam survives into the dryer it can dry as a resist film causing spots — check collapse by inspecting fabric immediately after application: a correct foam leaves a uniformly damp surface with no visible bubbles.
- **Restart and waste:** foam in pipes ages during stops; on restart, the aged foam must be purged, and shade/finish at the restart length checked.
- **Measurement burden:** the mill must measure density and WPU continuously — foam trades water cost for control cost.

8. Glossary — key terms

- **Blow ratio:** volume of foam produced per volume of liquor consumed.
- **Foam density:** mass per unit volume of foam; ≈ liquor density ÷ blow ratio.
- **Lamella:** the thin liquid film between two bubbles; Plateau borders are its drainage channels.
- **Foam half-life:** practical stability measure — time for half the liquid to drain or foam height to halve.
- **Wet pick-up (WPU):** liquor carried by fabric as % of dry mass.
- **Low wet pick-up (LWPU):** application family delivering finish at roughly 15–35% pick-up.

9. Key points

- Foam = air carrier + concentrated liquor; stability must last to the fabric, then collapse must be complete.
- Blow ratio sets foam density (10:1 on 1000 g/L liquor → 100 g/L); density × layer thickness fixes application.

- Cutting WPU from 70% to 25% cuts evaporation energy from 1582 to 565 kJ/kg — about 64% saved.
- Foam recipes are recalculated to deliver the same g/kg at low WPU; foam specifications belong on the recipe sheet.

10. Summary

Foam technology substitutes air for water as the application carrier. Surfactant films, controlled bubble size and stabilisers give a foam that survives transport; blow ratio and density metering give quantitative control; pad, knife and screen applicators lay it uniformly; and collapse on the fabric releases a concentrated liquor at 15–35% pick-up. The quantified prize is drying energy — roughly two-thirds of evaporation energy saved in the worked comparison — plus less migration, less hydrolysis and less effluent. The price is control: density, distribution, compatibility and collapse must all be engineered and monitored.

11. Practice

1. A liquor of density 1040 g/L is foamed at blow ratio 13:1. Calculate the foam density.
2. A line runs at 60% WPU conventionally and converts to foam at 20% WPU. Using latent heat 2260 kJ/kg, calculate evaporation energy per kg fabric in both cases and the percentage saving.
3. A pad recipe delivers softener at 30 g/L and 75% WPU. What liquor concentration is needed to deliver the same amount at 30% WPU?
4. Explain, using lamellae and Plateau borders, why raising liquor viscosity increases foam half-life.
5. State two causes of streaky foam finishing and one remedy for each.

Answer key

1. Foam density = 1040 / 13 = 80 g/L.
2. Conventional: 0.60 × 2260 = 1356 kJ/kg. Foam: 0.20 × 2260 = 452 kJ/kg. Saving = (1356 − 452)/1356 ≈ 66.7%.
3. Delivered = 30 × 0.75 = 22.5 g/kg. Required concentration = 22.5 / 0.30 = 75 g/L.
4. Drainage flows through Plateau borders out of the lamellae. Higher viscosity slows this flow, so lamellae stay thicker for longer, rupture is delayed, and both drainage half-life and foam lifetime increase (at the cost of slower spreading and penetration).
5. (Any two) Density drift across the manifold → balance feed paths and control in-line density; aged/coarsened foam after a stop → purge the line and check the restart length; defoaming contamination → test full-recipe compatibility and remove silicone/oil sources; uneven blade gap → set and verify gap across the width.